CURRICULUM VITAE ET STUDIORUM CHIARA GINI

CHIARA GINI, Ph.D.

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Scientific platforms: Google Scholar | LinkedIn | ResearchGate | ORCiD | Scopus | Github

Profile Postdoctoral researcher specializing in gut microbiome multi-omics, with expertise in shotgun metagenomics, metatranscriptomics, and bioinformatics pipeline development. Experienced in the integration of taxonomic and functional data layers across high-dimensional datasets, with a track record of international collaborations across Italy, Canada, and the United States. Committed to advancing precision phenomics and microbial ecology in livestock and environmental systems.

Professional experience

Postdoctoral research scholar
10/2024 – present | Department of Animal Science, North Carolina State University (NCSU), Raleigh, NC, USA | Supervisor: Prof. Christian Maltecca
Analytical & Technical Expertise:

- Shotgun metagenomics and metatranscriptomics analysis, including data processing, integration of taxonomic and functional layers (multi-omics), and interpretation of high-dimensional datasets (gut microbiome in sows)
- End-to-end bioinformatics workflows, data management, and scientific reporting (technical reports, preprints)

Tools & Languages: ATLAS*, HUMAnN, DIABLO, R, batch. Usage of HPC clusters (slurm, sbatch).

Post doctoral Fellow
04/2024 – 09/2024 | Microbial Ecology Group, Water Research Institute (IRSA), National Research Council of Italy (CNR), Verbania, Italy | Supervisor: Dr. Gianluca Corno
Analytical & Technical Expertise:

- Shotgun metagenomics analysis of microbial communities from marine environments, focused on analysis of biodiversity and antimicrobial resistance.

Tools & Languages: Kbase*, R.

Visiting International Research Student (VIRS)
09/2023 – 01/2024 | Centre for Genetic Improvement of Livestock (CGIL), Department of Animal Biosciences, University of Guelph, Ontario, Canada | Supervisor: Prof. Ángela Cánovas
Analytical & Technical Expertise:

- Amplicon sequencing and metatranscriptomics analysis of livestock-associated microbial communities

Tools & Languages: Qiime 1.9, Qiime 2, VSEARCH, MICCA, DADA2, CRCBio, R. Usage of HPC clusters.

Ph.D. researcher
11/2020 – 01/2024 | Department of Veterinary and Animal Science (DIVAS), University of Milan, Lodi, Italy | Supervisor: Prof. Fabrizio Ceciliani
Analytical & Technical Expertise:

- Amplicon sequencing analysis, benchmarking of bioinformatics pipelines, and biostatistical analysis of microbiome data**Tools & Languages:** Qiime 1.9, Qiime 2, VSEARCH, MICCA, DADA2, R. Usage of HPC clusters.

* ATLAS is a workflow and embeds all the tools dedicated to QC, Assembly, Binning, to Annotation, requiring the user to select the features adequately to its research. Similarly does KBase, which in turn is a platform and allows the computing directly online.

Graduate Student Researcher
01/2020 – 07/2020 | Department of Veterinary and Animal Science (DIVAS), University of Milan, Italy | Supervisor: Prof. Giulietta Minozzi

- Master's thesis research on genetic variability in the sex-determining locus (SDL/csd gene) in Italian honeybee populations (Apis mellifera)
- Developed a bioinformatics pipeline for allele discrimination from whole-genome NGS data; identified 20 unique csd alleles across 125 individuals sampled from 12 Italian regions

Undergraduate Student Trainée
10/2017 – 07/2018 | Department of Veterinary and Animal Science (DIVAS), University of Milan, Italy | Supervisor: Prof. Cristina Lecchi

- Bachelor's thesis research focused on microRNA expression analysis in formalin-fixed paraffin-embedded (FFPE) samples of canine cutaneous and oral melanoma using RT-qPCR

Education

Ph.D. in Veterinary and Animal Science
Università degli Studi di Milano (Italy) | 01/11/2020 - 01/02/2024. Title obtained on 16/09/2024 | *Supervisor*: Prof. Fabrizio Ceciliani

Advanced Course in Molecular Diagnostic Pathology (6ZT)
Università degli Studi di Milano (Italy) | 15/04/2022 - 14/10/2022

Master in Veterinary Biotechnological Sciences (LM-09)
Università degli Studi di Milano (Italy) | 10/2018 - 14/07/2020 | *Final score* 100/110 | *Supervisor:* Prof. Giulietta Minozzi

Bachelor in Biotechnology (K-09)
Università degli Studi di Milano (Italy) | 10/2014 - 23/10/2018 | *Final score* 90/110 | *Supervisor:* Prof.ssa Cristina Lecchi

Skills

Bioinformatics & Tools: ATLAS, HUMAnN, DIABLO, QIIME 1.9/2, DADA2, VSEARCH, MICCA, KBase, IGV, MegaX
Programming & Languages: R, Bash, Python
Platforms: HPC clusters (SLURM, SBATCH)
Other: Multi-omics data integration, pipeline benchmarking, scientific writing, technical reporting

Languages

Italian — Native
English — Fluent (2+ years based in the US)
Spanish — Intermediate (conversational)
French — Basic (B1 certification; limited current practice)

Grants & Awards

Travel Award – AGBT Agriculture Conference 2026
Issued by: The Animal Breeding-RCN: Farm Animal Genomics Collaborative | April 2026 Competitive travel grant (up to $3,000) awarded by the Animal Breeding-RCN Working Group based on the evaluation of a submitted research abstract and its contribution to Precision Phenomics in livestock genomics.

Scherago International Student Travel Grants Award
Issued by: Scherago International
One of six recipients of the Scherago International Student Travel Grant ($750), awarded for travel expenses and complimentary registration to attend the Plant and Animal Genome Conference (PAG).

Peer Review

Ad hoc reviewer for an international peer-reviewed journals in the life sciences.

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Congress Partecipations (Selected)

- Advances in Genome Biology and Technology Agricultural Meeting 2026 (AGBT Ag). Phoenix, Arizona, USA. 12-15 April 2026. Poster presentation.
- International Plant and Animal Genome Conference 2026 (PAG 33). San Diego, California, USA. 9-14 January 2026. Two oral presentations.
- Advances in Genome Biology and Technology Agricultural Meeting 2025 (AGBT Ag). Orlando, FL, USA. 31 March-2 April 2025. Oral presentation.
- 19th International Symposium on Microbial Ecology (ISME19). Cape Town, ZA, 18-23 August 2024. Poster presentation.
- International Plant and Animal Genome Conference 2024 (PAG 31). San Diego, California, USA. 12-17 January 2024. Oral presentation. Recipient of the Scherago International Student Travel Grant (Animal Section).
- 2023 Ontario Sheep Farmers Convention Annual General Meeting (AGM). Alliston, Ontario, Canada. 26-27 October 2023.
- 73rd Annual Meeting of the European Federation of Animal Science (EAAP). Porto, Portugal. 5-8 September 2022. Poster presentation

Teaching activities

- 12 hours of training in Cellular and Molecular Biology for the module Molecular Pathogenesis of Viral diseases, Course of Molecular Virology, for the students of the Master course of Veterinary Biotechnological Science, Academic year: 2019/2020
- Invited lecture at "The Animal Microbiome" course, held by Prof. Filippa Addis. 04/04/2023

Scientific publications

First-author publications:

- Gini C, Byrd MKH, Ahsin M, Ali L, Virginio Da Silva J, Correia de Almeida Regitano L, Van Vliet S, Brito LF, Johnson JS, Jiang J, Tiezzi F, Maltecca C. Physiological axes drive gut microbiome genomic, transcriptional, and metabolic functions in heat-stressed lactating sows. [Under revision - ISME Communication]
- Maltecca C, Gini C, Wen H, Byrd MKH, Van Vliet S, Brito FL, Johnson JS, Jiang J, Tiezzi F. Functional Guild Analysis Characterizes Gut Microbiome Responses to Heat Stress in Thermotolerant and Thermosensitive Lactating Sows Across Three Omics Layers. [Under revision - Journal of Animal Science and Biotechnology]
- Gini C, Farronato M, Manunza A, Ravasi G, Abbruscato P, Lorenzini EC, Stella A. Cross-kingdom dynamics of the subgingival bacteriome and mycobiome: a pilot study on the effects of a novel HA-H2O2-Glycine formulation to treat periodontitis. [Under revision - Journal of Dentistry]
- **Gini C**, Tiezzi F, Jiang J, Byrd MKH, Wen H, Johnson JS, Brito LF, van Vliet S, Maltecca C. Data-driven enterosignatures link gut microbiome reorganization to heat stress responses in lactating sows. Front. Microbiol. 17:1797687. doi: 10.3389/fmicb.2026.1797687
- Sabatino R†, **Gini C**†, Borgomaneiro G, Sbaffi T, Corno G, Sun K, Zhang XH, Di Cesare A. Sinking Particles as Repository of Antimicrobial Resistome Diversity in the Sansha Yongle Blue Hole. Sci Total Environ. 2026;1021:181585. doi: 10.1016/j.scitotenv.2026.181585
- Nehme R†, **Gini C**†, Vanbergue E, Even S, Biscarini F, Andrés S, Rault L, Noel F, Hardit V, Bouhallab S, Pereira DM, Ksouri R, Roussel P, López S, Cremonesi P, Castiglioni B, Caruso D, Faré F, Fontana M, Ceciliani F, Abdennebi-Najar L. The effects of Thymus capitatus essential oil topical application on milk quality: a systems biology approach. Sci Rep. 2025 Feb 7;15(1):4627. doi: 10.1038/s41598-025-88168-0.

Co-author publications:

- Bao Q, Wen H, Zeng L, Yang L, Wang J, He S, Yin H, Jiang Y, Qu X, Wang Z, Li X, Yang X, Teng J, Zhao P, Zhang D, Liu D, Cao G, Yu T, Ding R, Wang C, Zhang W, Tan Z, Zhu Y, Xia Y, Wang J, Zhao X, Tiezzi F, **Gini C**, Huang Y, See G, Schwab C, Xu R, Chen Z, Zhao Y, Xiang H, Zhou H, Ding X, Zhang Z, Tang Z, Li K, Maltecca C, Fang L, Jiang J, Yi G. Long-read sequencing reveals the impact of structural variation on gene expression and complex traits in pigs. Nat Genet. 2025. [Under revision - Nature Genetics].
- Bao Q, Zeng L, Wen H, Wang J, Yin H, Zhang Y, Liu L, Tiezzi F, See G, Schwab C, **Gini C**, Maltecca C, Li K, Fang L, Zhao Y, Jiang J. PIGSVOME: an integrated resource of porcine genomic structural variants for functional genomics and breeding applications. [Under revision - Journal of Genetics and Genomics]

- Morasi RM, Biscarini F, **Gini C**, Ceciliani F, Lecchi C, Salina A, Thaler Neto A, Langoni H, Gebara C, Crippa BL, Silva NC. Assessment of bacterial diversity in bovine bulk tank milk samples from Brazil. Braz J Microbiol. 2025 Dec;56(4):2507-20.
- de Oliveira Duarte FA, Ramos KK, **Gini C**, Morasi RM, Silva NC, Efraim P. Microbiological characterization of kombucha and biocellulose film produced with black tea and cocoa bean shell infusion. Food Res Int. 2024 Aug 1;190:114568.
- Andrés S, **Gini C**, Ceciliani F, Gutiérrez-Expósito D, Arteche-Villasol N, Martín A, Cremonesi P, Faré F, Ghaffari MH, Giráldez FJ, Abdennebi-Najar L. Essential oil supplementation in milk replacers: short- and long-term impacts on feed efficiency, the faecal microbiota and the plasma metabolome in dairy calves. J Dev Orig Health Dis. 2024 Apr 2;15:e5. doi: 10.1017/S2040174424000084.
- de Matos LG, da Silva Abreu AC, Alonso VP, Gonçalves JL, do Nascimento MD, Pflanzer Jr SB, Rezende-de-Souza JH, **Gini C**, Murad NF, Brandao MM, Silva NC. Comparison of bacterial diversity in wet-and dry-aged beef using traditional microbiology and next generation sequencing. The Microbe. 2024 Mar 1;2:100035.
- Ranilla MJ, Andrés S, **Gini C**, Biscarini F, Saro C, Martín A, Mateos I, López S, Giráldez FJ, Abdennebi-Najar L, Pereira D. Effects of Thymbra capitata essential oil on in vitro fermentation end-products and ruminal bacterial communities. Sci Rep. 2023 Mar 13;13(1):4153.
- Lecchi C, Zamarian V, **Gini C**, Avanzini C, Polloni A, Rota Nodari S, Ceciliani F. Salivary microRNAs are potential biomarkers for the accurate and precise identification of inflammatory response after tail docking and castration in piglets. J Anim Sci. 2020 May;98(5):skaa153.

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